

DHRUV KAURA

Ludhiana, Punjab, India | +91-9877027929 | dhruvkaura25@gmail.com
Github: github.com/DhruvKaura | LinkedIn: linkedin.com/in/dhruv-kaura/

PROFESSIONAL SUMMARY

Backend-focused Software Engineer specializing in building high-performance APIs and scalable systems using Python and FastAPI. Experienced in designing asynchronous services, optimizing backend performance, and developing production-ready applications. Strong foundation in system design, databases, and distributed systems.

EDUCATION

GNA UNIVERSITY

Cumulative GPA: 8.15/10

Phagwara, Punjab

Bachelors of Technology Computer Science & Engineering

2020-2024

WORK EXPERIENCE

PROTOLABZ ESERVICES

Phagwara, Punjab

Assistant Developer

April 2024-May 2026

- Designed and deployed RESTful APIs using Python and FastAPI serving 3+ production web applications, achieving ~35% reduction in average API response time via query optimization, connection pooling, and async I/O with asyncio.
- Built and maintained backend microservices integrated with WordPress and Shopify platforms, resolving 90%+ of API and CMS integration incidents within the same business day, minimizing client-facing downtime.
- Containerized multi-service infrastructure using Docker and Docker Compose (API, PostgreSQL, Redis), delivering consistent parity between development and production environments with zero environment-related release failures.
- Automated end-to-end CI/CD pipelines using GitHub Actions, incorporating lint checks, unit tests, and Alembic database migration workflows, cutting manual deployment effort by ~70% and eliminating regression releases.

A2IT

Mohali, Punjab

React.js Trainee

June 2023 - August 2023

- Independently designed, developed, and shipped a full-stack Car Service Website using ReactJS and Tailwind CSS within a 10-week program, applying component lifecycle, Context API, and client-side routing patterns.

CETPA INFOTECH PVT. LTD

Noida, Uttar Pradesh

Web Development Trainee

June 2022 - August 2022

- Built a responsive Coffee Shop Website UI in ReactJS using Hooks-based state management, demonstrating component-driven architecture and clean separation of concerns across an 8-week accelerated program.

PROJECTS

REAL-TIME API ORCHESTRATION ENGINE (OrchestrAI) ([Link](#))

- Architected a distributed AI task orchestration engine using Python and LLM-based agents, supporting 200+ concurrent API workflows with branching execution, fault tolerance, and sub-second scheduling latency.
- Improved pipeline execution performance by ~40% through migration from synchronous request handling to fully asynchronous processing using asyncio, coroutines, and non-blocking I/O.
- Developed modular YAML/JSON-driven workflow configurations, reducing new pipeline onboarding time from several hours to under 15 minutes while improving scalability and maintainability.

ADAPTIVE WORKFLOW INTELLIGENCE SYSTEM [\(Link\)](#)

- Built a production-grade asynchronous backend using FastAPI and PostgreSQL with JWT authentication, RBAC middleware, and full audit logging, supporting 5+ distinct workflow types.
- Integrated Redis and Celery for distributed background task processing with automatic retry on failure and real-time task status tracking, enabling decoupled, fault-tolerant job execution.
- Deployed complete infrastructure via Docker Compose; implemented GitHub Actions CI/CD with automated linting, test execution, and Alembic migration workflows, achieving zero failed deployments post-launch.

AI-POWERED SELF HEALING PIPELINE [\(Link\)](#)

- Engineered an AI-powered data ingestion pipeline using FastAPI, PostgreSQL, and Phi-3 (Ollama) that automatically detects schema drift, generates semantic column mappings, and self-heals incoming CSV datasets without manual intervention.
- Designed a scalable backend using Service-Repository architecture, SQLAlchemy ORM, and Alembic to manage dataset processing, workflow status tracking, and healing audit logs, ensuring maintainability and extensibility.
- Implemented production-ready engineering practices with Docker, GitHub Actions CI/CD, and Pytest for containerized deployment, automated testing, and continuous integration.

LEADERSHIP & ACTIVITIES

GNA UNIVERSITY COUNCIL OF INNOVATION (GUCI)

Phagwara, Punjab

Coding Head

2021-2022

- Organized 10+ technical events with 300+ participants from 5+ universities; managed logistics, problem-setting, and judging criteria for competitive programming rounds.

EDXCELLENCE

Remote

Python and Web Development Mentor

November 2025 - March 2026

- Mentored 30+ students in Python backend development and MERN STACK; 90%+ completed hands-on capstone projects, with multiple students securing internships post-program.

TECHNICAL SKILLS

Language: Python(primary), JavaScript (ES6+), SQL, TypeScript

Backend Frameworks: FastAPI, REST API Design, Node.js, Asyncio, Celery, SQLAlchemy, Alembic, Flask

Databases: PostgreSQL, MySQL, MongoDB, Redis (caching & message broker)

DevOps & Infrastructure: Docker, Docker Compose, GitHub Actions (CI/CD), Git, Postman, Linux

Auth & Security: JWT Authentication, OAuth2, Role-Based Access Control (RBAC)

Concepts: Microservices Architecture, Distributed Systems, Async Programming, System Design, API Rate Limiting, Connection Pooling, Caching Strategies

CS Fundamentals: Data Structures & Algorithms, DBMS, Operating Systems, Object-Oriented Programming (OOP)

CERTIFICATIONS & PUBLICATIONS

Certifications:

- Udemy Full Stack Development by Angela Yu
- AWS Academy Graduate- AWS Academy Cloud Foundation
- UpGrad Python Programming Bootcamp
- UpGrad Generative AI Foundations Certificate

Publications

- Fruits and Vegetable Sanitization and Washing Device(*Dhruv Kaura, Parul Parbhakar, Harjasleen Singh, Dr. Syed Mufassir Yaseen*)